

Solutions to "Algebraic Geometry and Arithmetic Curves"

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1 Some topics in commutative algebra

1.1 Tensor products

Exercise 1.1.1. Let $\{M_i\}_{i \in I}, \{N_j\}_{j \in J}$ be two families of modules over a ring A . Then

$$(\oplus_{i \in I} M_i) \otimes_A (\oplus_{j \in J} N_j) \cong \oplus_{(i,j) \in I \times J} (M_i \otimes_A N_j).$$

Proof. There is a chain of natural isomorphisms:

$$\begin{aligned} & (\oplus_{i \in I} M_i) \otimes_A (\oplus_{j \in J} N_j) \\ & \cong \oplus_{i \in I} (M_i \otimes_A (\oplus_{j \in J} N_j)) \\ & \cong \oplus_{j \in J} \oplus_{i \in I} (M_i \otimes_A N_j) \\ & \cong \oplus_{(i,j) \in I \times J} (M_i \otimes_A N_j). \end{aligned}$$

□

Exercise 1.1.2. There is a unique A -linear map

$$f : \text{Hom}_A(M, M') \otimes_A \text{Hom}_A(N, N') \rightarrow \text{Hom}_A(M \otimes_A N, M' \otimes_A N')$$

such that $f(u \otimes v) = u \otimes v$.

Proof. Consider the map $g : \text{Hom}_A(M, M') \times \text{Hom}_A(N, N') \rightarrow \text{Hom}_A(M \otimes_A N, M' \otimes_A N')$ sending (u, v) to $u \otimes v$. It is clearly bilinear, so factors uniquely through $\text{Hom}_A(M, M') \otimes_A \text{Hom}_A(N, N')$. □

Exercise 1.1.3. Let M, N be A -modules and let $i : M' \rightarrow M, j : N' \rightarrow N$ be submodules of M and N , respectively. Then there is a canonical isomorphism

$$(M/M') \otimes_A (N/N') \cong (M \otimes_A N) / (\text{Im } i_N + \text{Im } j_N).$$

In particular, $(\mathbf{Z}/n\mathbf{Z}) \otimes_{\mathbf{Z}} (\mathbf{Z}/m\mathbf{Z}) = \mathbf{Z}/l\mathbf{Z}$, where $l = \gcd(m, n)$.

Proof. Observe that

$$(M \otimes_A N)/(\text{Im } i_N + \text{Im } j_N) \cong ((M \otimes_A N)/(\text{Im } i_N))/(\text{Im } j_{M/M'}),$$

so Corollary 1.13 gives us a chain of isomorphisms

$$\begin{aligned} & ((M \otimes_A N)/(\text{Im } i_N))/(\text{Im } j_{M/M'}) \\ & \cong ((M/M') \otimes_A N)/(\text{Im } j_{M/M'}) \\ & \cong (M/M') \otimes_A (N/N'). \end{aligned}$$

For the last statement, just recall that $(m) + (n) = (l)$. □

Exercise 1.1.4. Let M, N be A -modules and let B, C be A -algebras.

- (a) If M and N are finitely generated over A , then so is $M \otimes_A N$.
- (b) If B and C are finitely generated over A , then so is $B \otimes_A C$.
- (c) The converses of (a) and (b) are in general false.

Proof. Parts (a) and (b) are obvious. For part (c), just notice that $\mathbf{Q} \otimes_{\mathbf{Z}} \mathbf{Z}/2\mathbf{Z}$ is the zero ring, but $\mathbf{Q}$ is not finitely generated over $\mathbf{Z}$. □

Exercise 1.1.5. Let $\rho : A \rightarrow B$ be a ring homomorphism, M an A -module, and N a B -module. There is a canonical isomorphism of A -modules

$$\text{Hom}_A(M, \rho_* N) \cong \text{Hom}_B(\rho^* M, N).$$

Proof. Let $f : \text{Hom}_A(M, \rho_* N) \rightarrow \text{Hom}_B(\rho^* M, N)$ be the map sending u to the map $f(u)$ s.t. $f(u)(m \otimes b) = bu(m)$. This is well-defined because if $n \otimes c = m \otimes b$, we may assume that there is some $a \in A$ s.t. $an = m$ and $\rho(a)b = c$ and so $f(u)(n \otimes c) = cu(n) = \rho(a)bu(n) = bu(m)$.

In addition, define $g : \text{Hom}_B(\rho^* M, N) \rightarrow \text{Hom}_A(M, \rho_* N)$ to be the map s.t. $g(u)(m) = u(m \otimes 1)$. This is immediately seen to be well-defined, and we contest that this is an inverse for f . Indeed, $f(g(u))(m \otimes b) = bg(u)(m) = bu(m \otimes 1) = u(m \otimes b)$ and $g(f(u))(m) = f(u)(m \otimes 1) = u(m)$. □

Exercise 1.1.6. Let $(N_i)_{i \in I}$ be a direct system of modules. Then for any A -module M , there exists a canonical isomorphism $\varinjlim (N_i \otimes_A M) \cong (\varinjlim N_i) \otimes_A M$.

Proof. Tensor-hom. □

Just kidding, it follows from the fact that $\varinjlim (N_i \times M)$ is canonically isomorphic to $(\varinjlim N_i) \times M$, and that a bilinear map over the latter must then induce a cocone of bilinear maps. □

Exercise 1.1.7. Let B be an A -algebra, and let M, N be B -modules. Then there exists a canonical surjection $M \otimes_A N \rightarrow M \otimes_B N$.

Proof. Do I really need to spell it out? □

1.2 Flatness

Exercise 1.2.1. Let M be an A -module and let $I \subseteq \text{Ann}(m)$ be an ideal.

- (a) There is a natural A/I -module structure on M , and $M \cong M \otimes_A A/I$.
- (b) Let N be another A -module such that $I \subseteq \text{Ann}(N)$. The canonical homomorphism $M \otimes_A N \rightarrow M \otimes_{A/I} N$ is an isomorphism.

Proof. For (a), observe that if $x \in I$, then $am = (a + x)m$ for any $a \in A$ and $m \in M$. This implies that the multiplication of elements of M by cosets $a + I$ is well defined, hence we have an A/I -module structure on M . To see that this is the extension of scalars, recall that $M \otimes_A A/I \cong M/(IM)$, which is just M .

Part (b) follows from a chain of isomorphisms

$$\begin{aligned} M \otimes_A N & \\ \cong (M \otimes_A A/I) \otimes_{A/I} N & \\ \cong M \otimes_{A/I} N, & \end{aligned}$$

where the first isomorphism comes from Corollary 1.11 and the second comes from part (a). $\square$

Exercise 1.2.2. Let $\rho : A \rightarrow B$ be a ring homomorphism, S a multiplicative subset of A , and $T = \rho(S)$. Then T is a multiplicative subset of B , and $T^{-1}B \cong B \otimes_A S^{-1}A$.

Proof. The first part is obvious. To see the last statement, let $\iota : B \rightarrow T^{-1}B$ be the canonical embedding. Since ι sends elements of $T = \rho(S)$ to units, there is a canonical homomorphism $\phi : S^{-1}A \rightarrow T^{-1}B$. Since $\otimes_A$ is a pushout in the category of A -algebras, there is a unique homomorphism $B \otimes_A S^{-1}A \rightarrow T^{-1}B$ such that this diagram commutes:

$$\begin{array}{ccc} S^{-1}A \otimes B & \longleftarrow & S^{-1}A \\ \uparrow & \searrow & \downarrow \\ B & \longrightarrow & T^{-1}B. \end{array}$$

This is an isomorphism because of universal property of localizations. $\square$

Exercise 1.2.3. Nakayama's lemma does not hold for modules that are not finitely generated.

Proof. Consider $\mathbf{Q}$ as a module over $\mathbf{Z}_{(p)}$. $\square$

1.3 Formal completion

2 General properties of schemes

2.1 Spectrum of a ring

Exercise 2.1.1. For k a field, $\text{Spec}(k[[T]])$ is the space $\{\xi, 0\}$ with generic point ξ and closed point 0.

Proof. It is easy to see that any power series with a non-zero constant term is a unit. Furthermore, if we have $f = \sum a_n T^n$ with $a_N = 1$ for some N and $a_n = 0$ whenever $n \leq N$, then we can pull out a common factor of T^N . We see that f is then a unit multiple of T^N and $(f) = (T^N)$. Clearly, this is only prime when $N = 1$. It is easy to see that $k[[T]]$ is an entire ring, so the proposition is immediate. $\square$

Exercise 2.1.2. Let $\varphi : A \rightarrow B$ be a homomorphism of finitely generated algebras over a field k . The image of a closed point under $\text{Spec } \varphi$ is a closed point.

Proof. If $\text{Spec } \varphi$ takes the maximal ideal $\mathfrak{m}$ into the ideal $\mathfrak{p}$, we get a diagram

$$k \rightarrow A/\mathfrak{p} \xrightarrow{\bar{\varphi}} B/\mathfrak{m}.$$

By Corollary 1.12, $B/\mathfrak{m}$ is a finite algebraic extension of k , so $B/\mathfrak{m}$ (resp. $A/\mathfrak{p}$) is finite over $A/\mathfrak{p}$ (resp. k). It then suffices to prove that $A/\mathfrak{p}$ is a field. Any $a \in A$ has an associated irreducible polynomial $P = \sum a_i X^i \in k[x]$ such that $P(a) = 0$. This polynomial must have a nonzero constant term, so we have

$$-\frac{1}{a_0} \sum_{i>0} a_i a^i = 1,$$

but the polynomial on the left has no constant term and we can pull out a factor of a . $\square$

Exercise 2.1.3. Let $k = \mathbf{R}$ be the field of real numbers and let $A = k[X, Y]/(X^2 + Y^2 + 1)$. Let x, y be the respective images of X, Y in A .

- (a) Let $\mathfrak{m}$ be a maximal ideal of A . Then there are $a, b, c, d \in k$ such that $x^2 + ax + b, y^2 + cy + d \in \mathfrak{m}$. Furthermore, $\mathfrak{m}$ is generated by a single element $f = \alpha x + \beta y + \gamma$ with at least one of α, β nonzero.
- (b) The map $(\alpha, \beta, \gamma) \mapsto (\alpha x + \beta y + \gamma)A$ establishes a bijection between the subset $\mathbf{P}(k^3) \setminus \{(0, 0, 1)\}$ of the projective space $\mathbf{P}(k^3)$ and the set of maximal ideals of A .
- (c) Let $\mathfrak{p}$ be a non-maximal prime ideal of A . Then the canonical homomorphism $\varphi : k[X] \rightarrow A$ is finite and injective, and $k[X] \cap \mathfrak{p} = 0$. Let $g \in \mathfrak{p}$ and let $g^n + a_{n-1}g^{n-1} + \dots + a_0 = 0$ be an integral equation with $a_i \in k[X]$. Then $a_0 = 0$, and so $\mathfrak{p} = 0$.

Proof. First, notice that $A/\mathfrak{m}$ is isomorphic to $\mathbf{C}$, since $X^2 + Y^2 + 1$ does not split in k . Then the existence of a, b, c, d follows from the fact that any complex number satisfies a monic quadratic equation. Since $\mathbf{C}$ is degree 2 over k , there are α, β, γ , not all zero, such that

$$\alpha x' + \beta y' + \gamma = 0,$$

where x' and y' are the respective images of x and y . Since not all of them are zero, we may assume that α is nonzero. Then we get an equation for x' in terms of y' , which we can then substitute into the equation $x'^2 + y'^2 + 1 = 0$. This new equation

cannot split in k because that would imply a real root of $x'^2 + y'^2 + 1 = 0$, so we have that $\mathfrak{m} = fA$, where f is the ideal corresponding to the linear equation in x' and y' .

Part (b) holds, since a solution to $X^2 + Y^2 + 1 = 0$ is determined (up to conjugacy) by an arbitrary complex number, which itself determines a linear equation with real coefficients.

That φ is finite and injective is clear, so consider the induced homomorphism $\bar{\varphi} : k[X]/\varphi^{-1}(\mathfrak{p}) \rightarrow A/\mathfrak{p}$. Notice that $B = k[X]/\varphi^{-1}(\mathfrak{p})$ is not a field, since $\bar{\varphi}$ is still finite and injective, which implies that $A/\mathfrak{p}$ is a field as well. Then $B = k[X]$ and $\mathfrak{p} \cap k[X] = 0$. Finally, in the integral equation for g , notice that a_0 must be a multiple of g , hence 0. Then the equation has a common factor of g which we may cancel, which gives a new equation with constant term a_1 . This repeats indefinitely and so $g = 0$. $\square$

Exercise 2.1.4. Let A be a ring.

- (a) Let $\mathfrak{p}$ be a minimal prime ideal of A . Then every element of $\mathfrak{p}A_{\mathfrak{p}}$ is nilpotent. In particular, every element of $\mathfrak{p}$ is a zero divisor.
- (b) If A is reduced, then any zero divisor in A is an element of a minimal prime ideal. This is not true in general if A is not reduced.

Proof. Part (a) is obvious.

For (b), let x be any zero divisor and let $\mathfrak{p}$ be a prime ideal minimal among those that contain x . If $\mathfrak{p}$ is minimal, we are done. Otherwise, suppose that $\mathfrak{q} \subset \mathfrak{p}$ is a minimal prime ideal not containing x . Since A is reduced, we must have $\sqrt{(x)} \cap \text{Ann}(x) = \{0\}$, so $\mathfrak{q}$ must contain a nonzero element $x' \in \text{Ann}(x)$. Localizing by $\mathfrak{q}$, we see that x is sent to 0 and x' is sent to a nilpotent. This implies that there is some positive integer n and some element $s \in A \setminus \mathfrak{q}$ such that $s(x'^n - x) = 0$, and it is easy to see that this means that $sx^2 = 0 \in \mathfrak{q}$, so $x^2 \in \mathfrak{q}$, contradiction. $\square$

The fact that this is false for A not reduced comes is immediate from the fact that there can be non-nilpotent zero divisors in a ring.

Exercise 2.1.5. Let k be a field. Let $\mathfrak{m}$ be a maximal ideal of $k[T_1, \dots, T_n]$.

- (a) Let $P_1(T_1)$ be a generator of $\mathfrak{m} \cap k[T_1]$ and $k_1 = k[T_1]/(P_1)$. Then we have an exact sequence

$$0 \rightarrow P_1 k[T_1, \dots, T_n] \rightarrow k[T_1, \dots, T_n] \rightarrow k_1[T_2, \dots, T_n] \rightarrow 0.$$

- (b) There exist n polynomials $P_1(T_1), P_2(T_1, T_2), \dots, P_n(T_1, \dots, T_n)$ such that $k[T_1, \dots, T_i] \cap \mathfrak{m} = (P_1, \dots, P_i)$ for all $i \leq n$. In particular, $\mathfrak{m}$ is generated by n elements.

Proof. Part (a) is obvious. In part (b), we proceed by induction. The statement is obvious in the single-variable case, so suppose that it holds for $n - 1$, with n an integer at least 2. Let k_1 be as in (a) and observe that it is a field, so the hypotheses and the exact sequence from (a) give a generating set $(P_2, \dots, P_n)$ for the image of $\mathfrak{m}$ in $k_1[T_2, \dots, T_n]$. This pulls back to a set of generators $(P_1, \dots, P_n)$ for $\mathfrak{m}$, and it is easy to see that the desired properties hold. $\square$

Exercise 2.1.6. Any closed subspace of a quasicompact space is itself quasicompact. If A is a ring, $\text{Spec}(A)$ is quasicompact.

Proof. Let X be a topological space and let $Y \subseteq X$ be any closed subset. If $\{U_i\}$ is an open covering of Y , $\{U_i \cup \complement_X Y\}$ is an open covering of X . Taking a finite subcovering of this extended covering yields a finite subcovering of the original covering.

For the second statement, suppose that $\{U_i\}$ is an open covering of $X = \text{Spec } A$. For each i , let J_i be the ideal such that $\complement_X U_i = V(J_i)$. We see that $\emptyset = \bigcap V(J_i) = V(\sum J_i)$, so there is a finite family of $f_k \in J_k$ such that $\sum f_k = 1$. Then $\sum J_k = 1$ and $\{U_k\}$ is a finite subcovering. $\square$

The next exercise is a bit open ended in character, so the proof label is changed to reflect that.

Exercise 2.1.7. Describe the closed subsets of $\text{Spec}(\mathbb{Z}[T])$. Show that they are not all principal.

Answer. The closed subsets of $\text{Spec}(\mathbb{Z}[T])$ can be broken up into three families: points, lines, and the generic point. The points are simply maximal ideals; in $\mathbb{Z}[T]$ these are ideals of the form (p, P) , where p is a non-zero prime number and P is an irreducible polynomial. The lines are just principal ideals; I call them lines because they determine a family of points that looks kind of like a line. Finally, there is the generic point (0) . The fact that the closed subsets are not all principal follows from the fact that the maximal ideals have 2 generators. $\square$

2.2 Ringed topological spaces

Let X be a topological space.

Exercise 2.2.1. Let A be a non-trivial abelian group and let $\mathcal{A}_X$ be the constant presheaf of A on X . The associated sheaf $\mathcal{A}_X^1$ is the sheaf of locally constant functions valued in A . In particular, $\mathcal{A}_X$ is a sheaf if and only if every non-empty open subset of X is connected.

Proof. Let $\mathcal{F}$ be the sheaf of locally constant A -valued functions on X and let $\varphi : \mathcal{A}_X \rightarrow \mathcal{F}$ be the evident morphism. It is easy to see that the stalks of $\mathcal{F}$ are all A . Let $\mathcal{G}$ be any sheaf and let $\alpha : \mathcal{A}_X \rightarrow \mathcal{G}$ be any morphism. If f is a locally constant function on U , we may split U into disjoint open sets $U_a = f^{-1}(a)$ for $a \in f(U)$. We define $\tilde{\alpha}(U) : \mathcal{F} \rightarrow \mathcal{G}$ to be the map sending f to the regluing of

the sections $\alpha(U)(a)|_{U_a}$. It is easy to see that this defines a morphism of sheaves and that $\tilde{\alpha} \circ \varphi = \alpha$. Since φ is surjective, $\tilde{\alpha}$ is the unique such morphism. $\square$

Exercise 2.2.2. Let $\mathcal{F}$ be a sheaf on X . Let $s, t \in \mathcal{F}(X)$. Then the set of $x \in X$ such that $s_x = t_x$ is open in X .

Proof. If $s_x = t_x$ at some point $x \in X$, there must be a neighborhood U_x of x such that $s_y = t_y$ for all $y \in U_x$. Taking the union of all such U_x gives the result. $\square$